

BIB Special session — Optogenetics: a light switch for nerve cells

- Optogenetik: sinir hücrelerini ışıqla açıp kapatmak
- The 2026 Nobel Prize in Physiology or Medicine, announced 5 October 2026
- Yasin Kaymaz, PhD



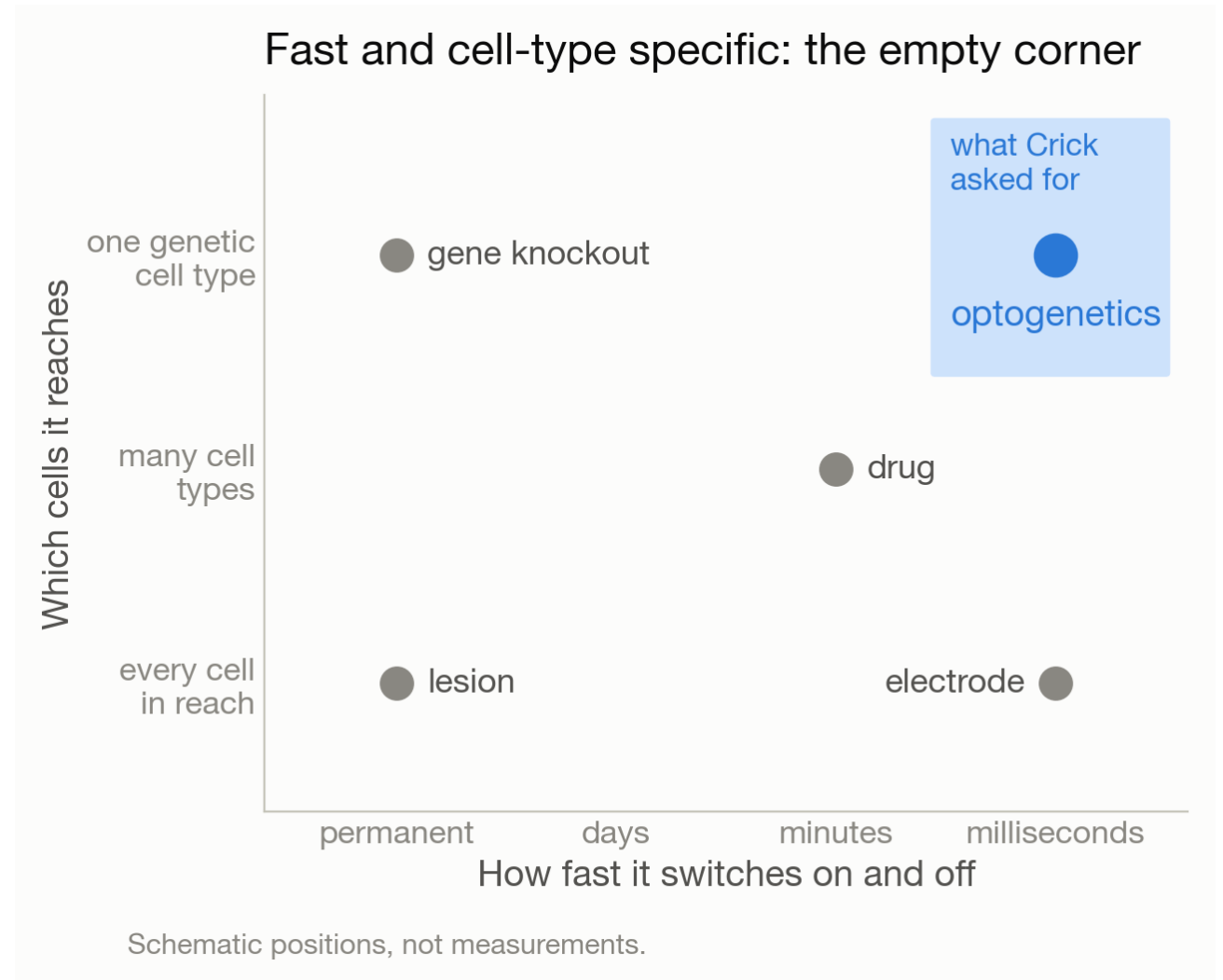
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The 2026 Nobel Prize in Physiology or Medicine

- "For their discoveries concerning light-gated ion channels and optogenetics"
- **Peter Hegemann**, Humboldt University of Berlin: found the alga's light-sensor genes
- **Georg Nagel**, University of Würzburg: showed they make light-gated ion channels
- **Karl Deisseroth**, Stanford University and HHMI: made those channels switch neurons on
- The committee's headline: light-seeking algae gave us a switch for nerve cells
- Announced 5 October 2026; 12 million Swedish kronor, shared equally

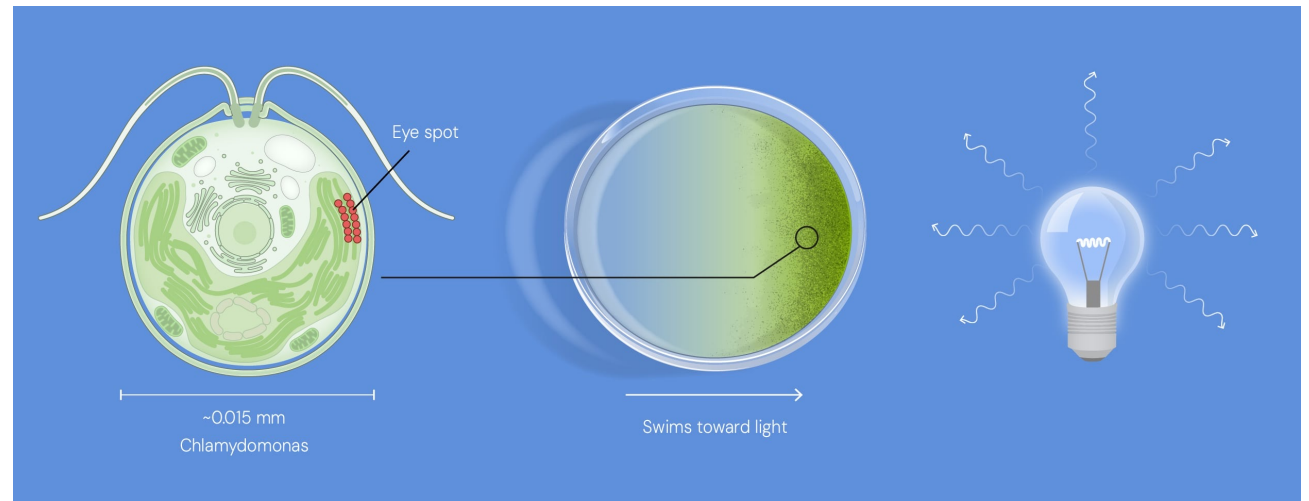
The question the old tools could not answer

- 20th-century brain maps: correlation, not cause
- Electrodes: fast, but every cell near the tip fires
- Drugs, lesions, knockouts: minutes, days or for good
- Crick's wish: one cell type, at nerve-signal speed, maybe with light
- So: is this cell type sufficient? Is it necessary?



It began with a question about an alga

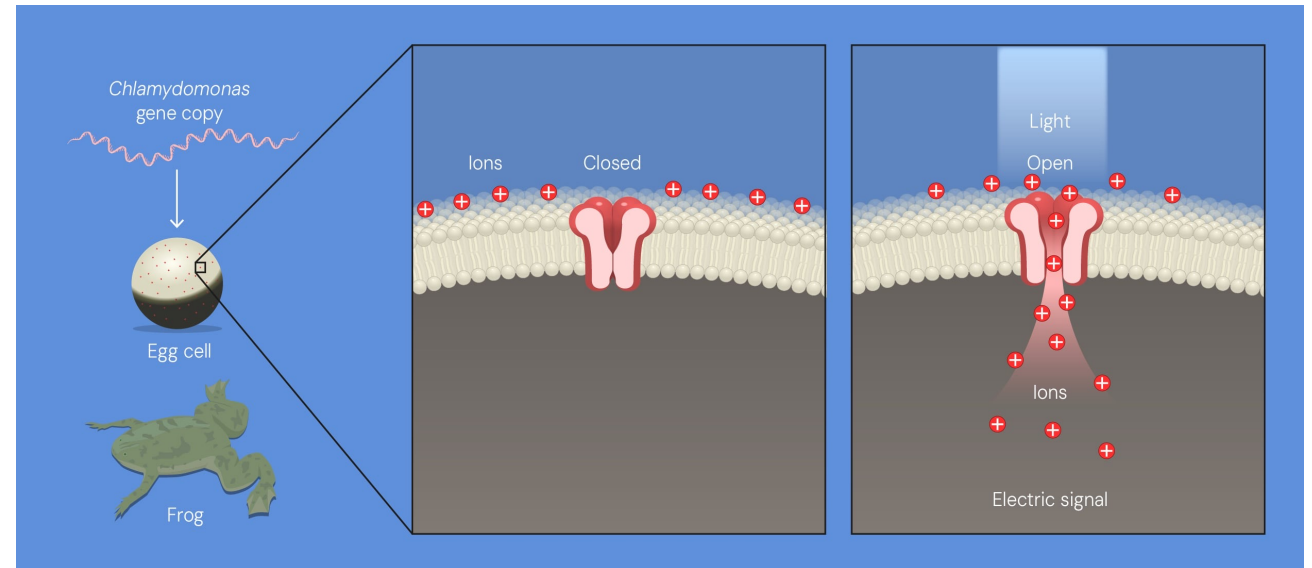
- Chlamydomonas swims toward light, sensed by its eyespot (göz lekesi)
- Its current starts 0.5 ms after a flash: 20× faster than our eyes
- Hegemann's guess: one protein senses the light and is the channel
- ~2000: two genes in a public algal gene collection looked like light sensors
- That was a similarity search (Week 4)



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Channelrhodopsin: a light-gated ion channel

- Nagel put each gene into frog eggs: light opened the new channels
- Channelrhodopsin-2 (ChR2) opens in 0.2 ms; cations flow in
- It worked in every cell type they tried: a portable tool (2003)
- Gated by ligand, by voltage, and now by light (ışık kapılı iyon kanalı)
- Later, an off switch: yellow light drives a chloride pump



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Video: optogenetics in four minutes

VIDEO · 4:26

Method of the Year 2010: Optogenetics

Nature Video, December 2010

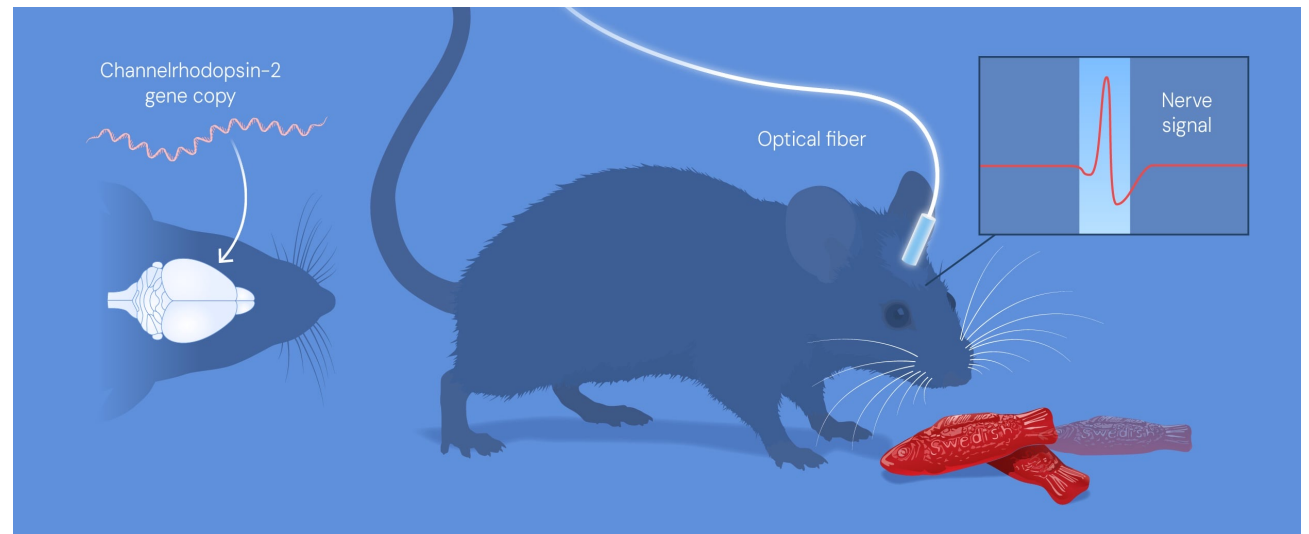


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[Nature Video, "Method of the Year 2010: Optogenetics" \(4:26\).](https://youtu.be/l64X7vHSHOE)

From a dish (2005) to a living mouse (2007)

- Deisseroth asked Nagel for the gene and put it into cultured rat neurons
- 2005: flashes of blue light made them fire, spike by spike
- 2007: ChR2 in one cell type of mouse motor cortex; light through an optical fibre (optik fiber) moved the whiskers
- The recipe: gene + cell-type promoter + virus + light



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Video: blue light on, the mouse runs

VIDEO · 0:28

Blue light on, the mouse runs

Channelrhodopsin-2 in the motor cortex
Deisseroth lab, Stanford



Click to play · streams from YouTube · youtu.be/v7uRFVR9BPU

[Channelrhodopsin-2 in the mouse motor cortex, blue light through the fibre. Deisseroth lab, Stanford \(0:28\).](https://youtu.be/v7uRFVR9BPU)

Video: one circuit, one behaviour

VIDEO · 0:49

Switching on hunting

Central amygdala neurons, mouse
New Scientist · Han et al., Cell 2017



Click to play · streams from YouTube · youtu.be/42bLSZnVf0g?t=6

[Hunting: central amygdala neurons switched on by light \(New Scientist; Han et al., Cell 2017\). 0:49](https://youtu.be/42bLSZnVf0g?t=6)

VIDEO · 0:58

Switching on eating

Extended amygdala to hypothalamus, mouse
Jennings et al., Science 2013

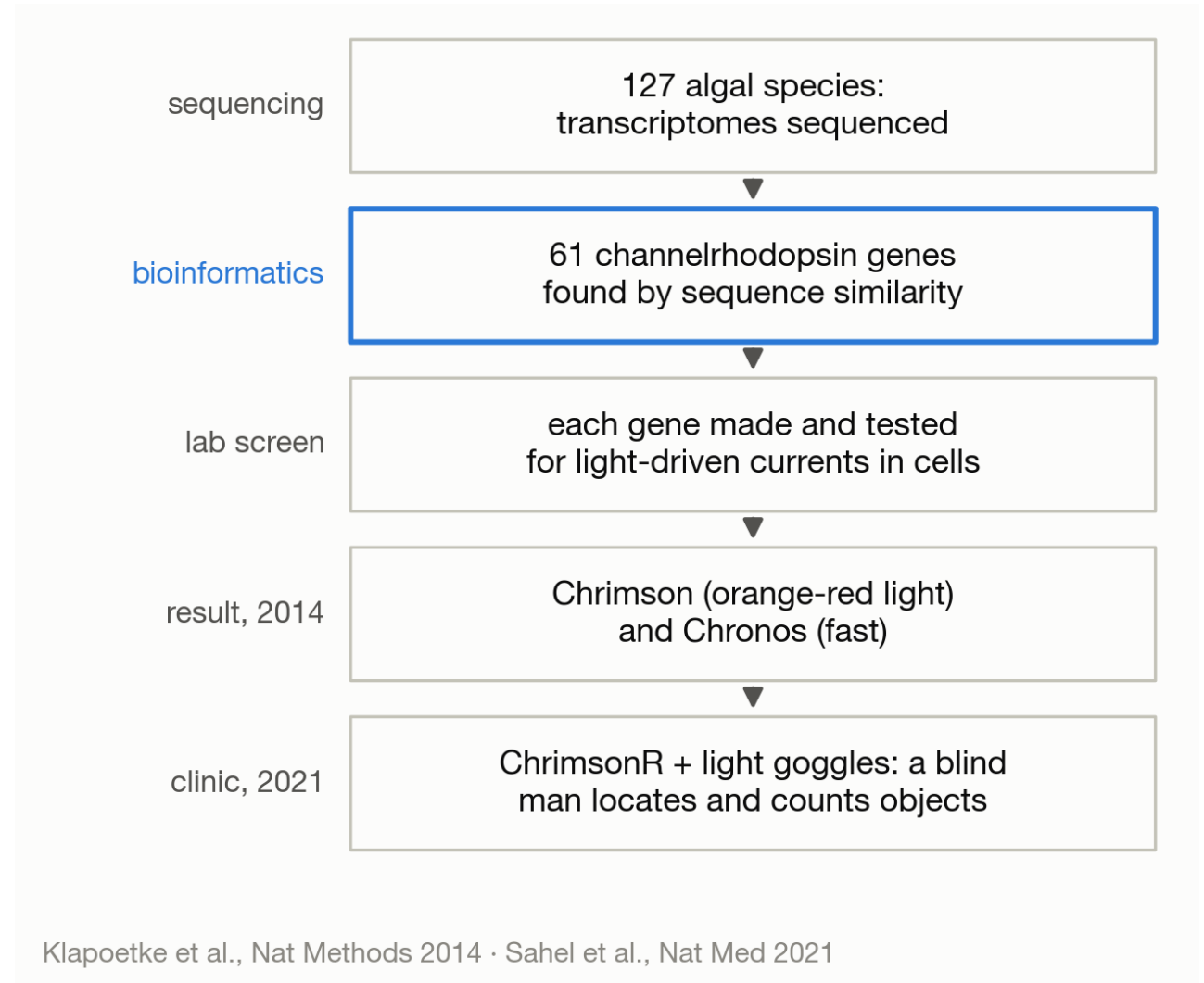


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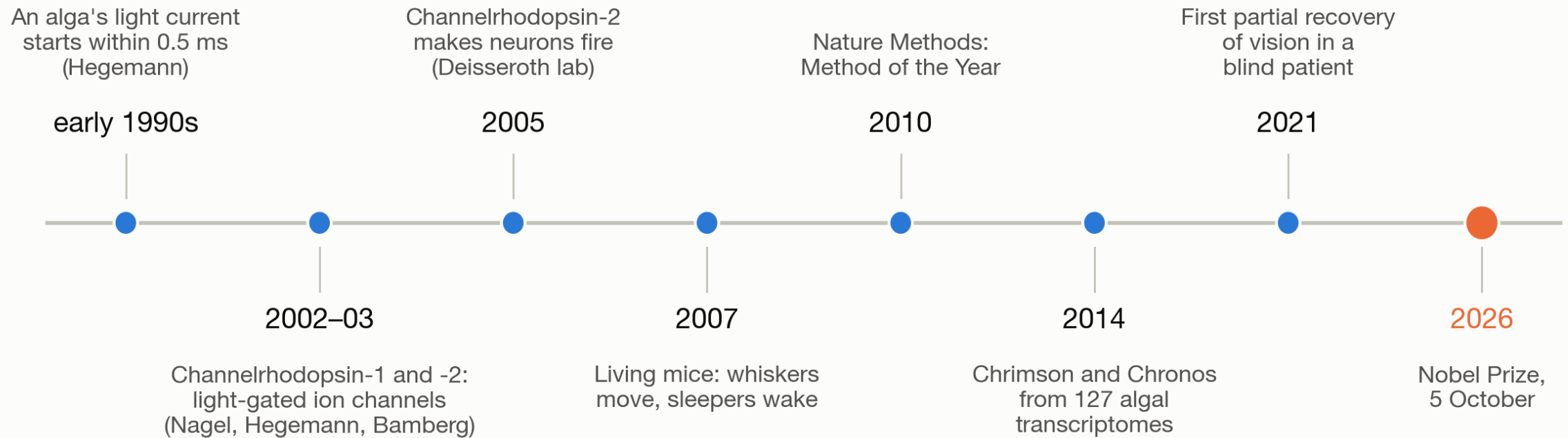
[Eating: an extended amygdala to hypothalamus pathway switched on by light \(Jennings et al., Science 2013\). 0:58](https://youtu.be/7Mmsah0v9Qc)

From 127 algal transcriptomes to a patient's eye

- 2014: 127 algal transcriptomes → 61 channelrhodopsins, each tested in cells
- Winners: Chrimson (orange-red light), Chronos (fast)
- 2021: ChrimsonR into the retina of a man blind from retinitis pigmentosa
- With light-projecting goggles he could locate, count and touch objects
- One patient, partial vision: a start, not a cure



From an alga to a Nobel Prize



Events evenly spaced; the years are not to scale.

Take-aways, and one question for you

- Optogenetics = a light-gated channel + genetic targeting + light delivery
- Genes decide which cells; light decides when, to the millisecond
- It turned brain maps from correlation into cause
- It began with curiosity about an alga, and a sequence search
- Three laureates, many more parents: Bamberg, Boyden, Zhang, Miesenböck ...
- Your turn: you may switch ONE cell type on or off. Which, and what do you measure?

APPENDIX (not taught — posted to EgeDers): read and watch more

- Nobel popular background (pdf): nobelprize.org/uploads/2026/10/popular-medicineprize2026.pdf
- Nobel scientific background (pdf): nobelprize.org/uploads/2026/10/advanced-medicineprize2026.pdf
- Nagel et al. 2002, Science (channelrhodopsin-1): doi:10.1126/science.1072068
- Nagel et al. 2003, PNAS (channelrhodopsin-2): doi:10.1073/pnas.1936192100
- Boyden et al. 2005, Nature Neuroscience (neurons in a dish): doi:10.1038/nn1525
- Aravanis et al. 2007, Journal of Neural Engineering (living rodents): doi:10.1088/1741-2560/4/3/S02
- Klapoetke et al. 2014, Nature Methods (Chrimson, Chronos): doi:10.1038/nmeth.2836
- Sahel et al. 2021, Nature Medicine (vision): doi:10.1038/s41591-021-01351-4
- The two behaviour videos: Han et al. 2017, doi:10.1016/j.cell.2016.12.027 · Jennings et al. 2013, doi:10.1126/science.1241812
- More video, 1:37: "Can mice be friends?" (Northwestern University): youtu.be/XNxcHmQVf2A
- More video, 10:18: fibre-optic implantation (neuroleap): youtu.be/kchWb1rC1iE